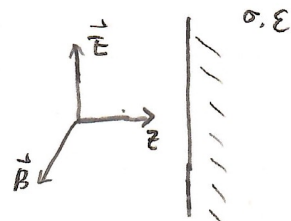


7.4 (a) The wave solution in the free space is given by

$$\vec{E}(\vec{r}, t) = \vec{E} (E_i e^{i(kz - \omega t)} + E_r e^{-i(kz + \omega t)})$$



Inside the medium, following 5.18, the wave solution is

$$\vec{E}(\vec{r}, t) = \vec{E} E_t e^{-z/\delta} e^{i z/\delta - i \omega t},$$

where $\delta = \sqrt{2/\mu\omega\sigma}$. The boundary condition at $z=0$ is given by

$$E_i + E_r = E_t$$

$$k(E_i - E_r) = \frac{1+i}{\delta} E_t$$

These equations lead to

$$E_t = \frac{2}{1 + \frac{1+i}{k\delta}} E_i, \quad E_r = \frac{1 - \frac{1+i}{k\delta}}{1 + \frac{1+i}{k\delta}} E_i$$

For the reflected wave, $E_r = \frac{(k\delta - 1) - i}{(k\delta + 1) + i} E_i = \frac{k^2\delta^2 - 2 - i \cdot 2k\delta}{(k\delta + 1)^2 + 1}$

$$\left| \frac{E_r}{E_i} \right| = \left[\frac{(k\delta - 1)^2 + 1}{(k\delta + 1)^2 + 1} \right]^{1/2}, \quad \arg\left(\frac{E_r}{E_i}\right) = -\arctan\left(\frac{2k\delta}{k^2\delta^2 - 2}\right)$$

(b)

$$R = \left| \frac{E_r}{E_i} \right|^2 = \frac{k^2\delta^2 - 2k\delta + 2}{k^2\delta^2 + 2k\delta + 2}$$

For good conductor, $k\delta = \frac{\omega}{c}\delta \ll 1$.

$$R = \frac{1 - k\delta + k^2\delta^2/2}{1 + k\delta + k^2\delta^2/2} = (1 - k\delta + k^2\delta^2/2)(1 - k\delta + k^2\delta^2/2)$$

$$= 1 - 2k\delta = 1 - 2\frac{\omega}{c}\delta$$